

WonSeok Choi 최원석

BASIC INFORMATION

First Name	WonSeok (원석)	Email	choidol0208@gmail.com
Family Name	Choi (최)	LinkedIn	linkedin.com/in/choi-wonseok
Country	Republic of Korea	Instagram	@namedchoi

EDUCATION & AFFILIATION

Korea Aerospace University	2019 – Present
- Undergraduate Student, School of Aerospace and Mechanical Engineering (Aerospace Engineering) - Undergraduate Researcher, IDOL Lab (idol.kau.ac.kr)	

RESEARCH INTERESTS

AI-driven design automation for rocket propulsion and composite structures, with a focus on reinforcement learning

SELECTED EXPERIENCE

Society of Rocket Study (SRS), Student Rocketry Club at Korea Aerospace University	
Member	2019 – Present
Technical Advisor	2026 – Present
President	2020
Noshiro Space Event, Japan (SRS)	
Technical Advisor	2026
Deputy Project Manager & Structures Team Lead	2025
Polaris-1 Liquid Rocket Project (SRS)	
Deputy Project Manager & Injector Lead	2023 – 2024
NURA (Korea's Inter-University Rocketry Association)	
Avionics Development Project with Korean Air	2022 – 2023
Deputy Lead; Configuration Manager	
Republic of Korea Army	
Mandatory Military Service	2021 – 2022

PUBLICATIONS

- H. Park, **W. Choi**, S. Han, et al., "Current Development Status of the Polaris-1 Small Liquid Rocket," *Journal of the Korean Society of Propulsion Engineers*, vol. 28, no. 6, pp. 62–73, 2024. [doi:10.6108/KSPE.2024.28.6.062](https://doi.org/10.6108/KSPE.2024.28.6.062) (in Korean)
- W. Choi**, S. Han, K. Park, et al., "Current Development Status of the Polaris-1 Small Liquid Rocket," *KSPE Spring Conference*, Jeju, Korea, May 2024. (in Korean)
- W. Choi**, H. Lee, H. Park, et al., "Basic Design of Ethanol/H₂O₂ Small Liquid Rocket Engine," *KSPE Spring Conference*, Jeju, Korea, May 2023. (in Korean)

CONFERENCE PRESENTATIONS

- W. Choi** and S. Kang, "Reinforcement Learning-Based Automated Layup Design of Recycled CFRP Under Material Uncertainty," *MRS Spring Meeting & Exhibit*, Honolulu, HI, USA, Apr. 2026. (Poster)
- W. Choi**, J. Kim, and S. Kang, "Design Automation of Recycled Carbon Fiber-Reinforced Polymer via Reinforcement Learning," *KSME 80th Anniversary Conference*, Jeongseon, Korea, Dec. 2025. (Oral, in Korean)

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